



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/02

Paper 2

April 2025

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **16** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme.
Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)(i)	kidney(s)	1	
1(a)(ii)	urine	1	Accept urea (and water)

Question	Answer	Marks	Further Information
1(b)	(strength) shows filtration or removes some impurities (limitation) (idea that) does not show liquid being stored in the bladder or liquid is continually leaving the bladder or not a continual process or water has to be poured into the kidneys or impurities remain in the bottles or (idea that) it does not show the entire renal system or (idea that) it does not show the waste products present in urine or it shows clean water (not waste products) or (idea that) it is not clear if the water that goes in and leaves is blood or (idea that) not to scale	2	correct strength = 1 mark correct limitation = 1 mark Ignore represents the renal system accurately Ignore cleans the water / strains the blood Accept ora for limitations Accept in kidneys the impurities leave as urine Accept higher level ideas such as, does not show reabsorption, artery and vein, cortex and medulla, Bowman's capsule, loop of Henle etc. Ignore kidneys do not have rocks and sand

Question	Answer	Marks	Further Information
1(c)	new evidence or more understanding	1	Accept technology advances or correcting mistakes in the model or scientific research gets better

Question	Answer	Marks	Further Information
2(a)	electrons	1	Accept shared pair of electrons Ignore covalent bond
2(b)	bromine is in same <u>group</u> (of the Periodic Table) as chlorine	1	Accept bromine and chlorine have the same number of electrons in their outer shells Accept bromine is also in Group 7 or both are in Group 7 or both are in Group 17 Note bromine is in Group 7 is not sufficient or bromine has seven electrons in its outer shell is not sufficient the answer must imply both elements are in the same group
2(c)	gains (one) electron(s)	1	Accept e ⁻ or e to represent an electron Do not accept shares electrons Do not accept gains electrons from another chlorine atom but accept gains electrons from another atom

Question	Answer	Marks	Further Information
3(a)	copper	1	Accept Cu
3(b)	M1 correct labels for both axes with units on y-axis M2 suitable scale for the y-axis and correct arrangement of bars M3 correct bars	3	Accept (labels) metals and each metal labelled and time (taken for pin to fall) in seconds = 1 mark Accept any order for the metals Accept suitable scale for y-axis that ensures the biggest bar occupies approximately half of the graph paper and bars are the same width with a tolerance of 1 square and separated by a gap = 1 mark Note the gap between bars does not have to be equal correct height of bars with tolerance of $\pm$ half a small square = 1 mark Note a line graph can only be awarded M1 Note M3 dependent on a scale for the y-axis

Question	Answer	Marks	Further Information
4(a)	any two from (idea of) sea level rising flooding droughts extreme weather conditions	2	Accept polar icecaps melting or glaciers melting or ocean temperature rising Note ice melting is not sufficient Accept desertification Accept named extreme conditions, e.g. (more) tornadoes or (more) hurricanes or unpredictable/less rainfall or unpredictable weather or rise in (global) temperatures or global warming or hotter weather Ignore pollution or heat stress Note wildfires is not sufficient but accept more wildfires Accept habitat destruction or deforestation or change in habitat Accept (idea of) extinction of species Accept lack of food or lack of (clean) water Note more natural disasters is not sufficient
4(b)	respiration or combustion or decomposition	1	Accept burning or decay Ignore breathing

Question	Answer	Marks	Further Information
5(a)	gametes	1	Accept sperm and egg/ovum/ova Ignore sex cells or reproductive cells
5(b)	genes (made of) DNA	1	Both correct for the mark Accept genetic information or genetic material for genes Note genetic is not sufficient Accept deoxyribonucleic acid
5(c)	50% or 1 in 2 XX are girls	2	each correct answer = 1 mark Accept $\frac{1}{2}$ or 0.5 or $\frac{2}{4}$ or equal chance or 'fifty-fifty' or evens Accept XY are boys Note two XX and/or two XY is not sufficient the answer must clearly state that XX is a girl or XY a boy

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Question	Answer	Marks	Further Information																
6(a)	table drawn with column or row headings e.g. metal or block and height and length and width cm units in headings for height and length and width all the data correctly entered into table	3	each correct answer = 1 mark Accept h for height, l for length and w for width Do not accept units written in the body of the table example table <table border="1"> <thead> <tr> <th>metal block</th><th>height in cm</th><th>length in cm</th><th>width in cm</th></tr> </thead> <tbody> <tr> <td>A</td><td>3</td><td>3</td><td>2</td></tr> <tr> <td>B</td><td>2.5</td><td>1.5</td><td>3</td></tr> <tr> <td>C</td><td>4</td><td>3</td><td>4</td></tr> </tbody> </table> Ignore extra columns or rows e.g. volume	metal block	height in cm	length in cm	width in cm	A	3	3	2	B	2.5	1.5	3	C	4	3	4
metal block	height in cm	length in cm	width in cm																
A	3	3	2																
B	2.5	1.5	3																
C	4	3	4																
6(b)	(density =) mass ÷ volume or $88 \div 11$ (density =) 8 (unit) g/cm³	3	8 g/cm³ with or without working = 3 marks 8 and no unit or incorrect unit = 2 marks correct equation or evidence of correct substitution into equation = 1 mark Note equation takes precedence over correct substitution correct unit = 1 mark Accept appropriate answers with kg/m³ or kg/cm³ Note answer on answer line takes precedence for density and unit																
6(c)	E followed by D followed by F	1	all three in correct order for the mark Accept 21.4, 16.3, 5.2																

Question	Answer	Marks	Further Information
7(a)	80 (J)	1	Note answer on answer line takes precedence
7(b)	thermal	1	Accept heat Do not accept light energy
7(c)	$\frac{\text{useful energy}}{\text{electrical energy}} \times 100$ or $\frac{120}{200} \times 100$ 60	2	60 with or without working = 2 marks correct equation or evidence of correct substitution = 1 mark Note equation takes precedence over correct substitution Accept 0.6 = 1 mark if no other marks scored

Question	Answer	Marks	Further Information
8(a)	increasing the mass of the marble will increase the diameter of the crater	1	Accept the heavier the marble the bigger the crater or greater mass gives a bigger hole Accept ora Accept mass of marble is proportional to the diameter of the crater Ignore reference to size of marble
8(b)	any one from depth of sand type of sand material of marble diameter of marble	1	Accept amount of sand or mass of sand or level of sand (in container) or volume of sand Ignore size of container of sand Note container of sand is not sufficient Accept density of marble Accept volume of marble or size of marble or type of marble Ignore height of drop or force of dropping marble Do not accept mass of marble

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Question	Answer	Marks	Further Information
8(c)	any two from (idea of mass) extinction climate change destruction of cities	2	each correct answer = 1 mark Accept death of many species or animal extinction Ignore many deaths Accept global cooling / ice-age Ignore global warming Accept destruction of infrastructure Accept as extra marking points • a huge crater or shockwave or wildfires • tidal waves or flooding • (huge) dust clouds or reduction of sunlight • loss of habitat • disruption of food chains Ignore earthquake or explosions or forming a new moon

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Question	Answer	Marks	Further Information
9(a)	colour of feathers or markings of feathers or beak shape or beak markings or beak colour or colour of feet	1	Accept eye colour or fur colour or sex of puffin Ignore skin colour or eye shape Do not accept size of beak or size of bird Note unqualified colour or beak is not sufficient Note type of feet is not sufficient
9(b)(i)	increases	1	
9(b)(ii)	population decreases and one from become extinct endangered starves migrate to another area lack of food source	2	each correct answer = 1 mark Accept (idea that) population stays the same as puffins eat alternative food source = 2 marks

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Question	Answer	Marks	Further Information
10(a)	98 (°C)	1	Accept any value between 70 °C and 140 °C
10(b)(i)	any one from do the demonstration behind safety glass or in a fume cupboard wear (safety) goggles or laboratory coat	1	Ignore stays away from the experiment Accept use a small sample of sodium or use tweezers to handle sodium Accept wears (safety) glasses Ignore wears gloves / safety gear
10(b)(ii)	any one from moves more quickly the reaction happens more quickly burns with a lilac flame or burns with a pink flame	1	Note the answer must be comparative Note assume answer refers to potassium, but accept ora if answer specifies sodium Accept potassium is more reactive or potassium has a more violent reaction or a more vigorous reaction Accept does not burn with a yellow/orange flame or burns with a different coloured flame or burns with a bigger flame

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Question	Answer	Marks	Further Information
10(b)(iii)	potassium + water $\longrightarrow$ potassium hydroxide + hydrogen	1	Accept reactants in either order and products in either order Accept correct formulae and ignore balancing, but names takes precedence, e.g. $2K + 2H_2O \longrightarrow 2KOH + H_2$ Accept = for $\longrightarrow$
Question	Answer	Marks	Further Information
11(a)(i)	A and C	1	both required in either order
11(a)(ii)	A and F	1	both required in either order
11(b)(i)		2	M1 correct position of peaks and troughs = 1 mark M2 correct amplitude = 1 mark Note M2 is dependent on M1
11(b)(ii)		1	Note peaks and troughs must be in the positions as shown in diagram but accept any height or any amplitude

Question	Answer	Marks	Further Information
12	dust gas (stellar) nurseries	3	each correct answer = 1 mark Accept dust and gas in either order Accept hydrogen or helium for gas